

Do BERT Models Generalise German Plurals Like Humans? Evidence from Distributional and Uncertainty Analyses

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KSANLPD1KU – Advanced NLP & Deep Learning – Autumn 2025

Abstract

German plural formation is a challenging test case for morphological generalisation because multiple plural classes compete and uncertainty increases for novel forms. Most NLP evaluations of plural prediction emphasise top-1 accuracy, but this can obscure how models behave under ambiguity and how closely their predictions resemble human response distributions. We compare human plural-choice distributions for German real, alternation, and nonce nouns to predictions from a trigram baseline, a monolingual German BERT (gBERT), and multilingual BERT (mBERT). We evaluate models using Jensen–Shannon distance (JSD) to human response distributions and uncertainty metrics (entropy, varentropy, and entropy-derived confidence), supplemented by statistical tests and qualitative error analysis. Humans show a strong uncertainty shift from real to nonce items, while BERT models show weaker condition sensitivity; the trigram baseline is unexpectedly competitive, and many gBERT deviations are high-confidence and consistent with local subword/lexical attraction. These results suggest that subword Transformers capture some regularities in German plural formation but rely on cue-use strategies that differ from human uncertainty calibration and cue integration.

1 Introduction

Morphological generalisation remains a core challenge for NLP systems, especially under out-of-vocabulary (OOV) conditions where memorisation is not sufficient. This challenge is particularly visible in inflectional morphology, where the correct output can depend on interacting phonological, lexical, and grammatical cues. Subword-based Transformer models are often assumed to improve robustness to OOV forms because they can compose representations from smaller units, but subword coverage does not necessarily imply human-like generalisation. While Transformer-based models achieve

state-of-the-art performance on high-resource data, they often struggle to generalise morphosyntactic rules to unseen stems (Liu and Hulden, 2021).

German plural formation provides a useful test case for this question. German plural marking is a notoriously complex system, with several competing suffix classes $-(e)n$, $-(e)r$, $-s$, $-e$, $-\emptyset$, entrenched gender-plural syncretism (e.g., the strong association between feminine nouns and $-n$), and the influence of prosodic constraints. Umlaut of the stem vowel may co-occur with suffixation or appear alone. In addition, multiple plural variants may be acceptable for some nouns, which means that plural choice is often better represented as a distribution over options than as a single deterministic label.

This makes German pluralisation especially suitable for a distributional human–model comparison. Rather than asking whether a model predicts the one “correct” plural, we can ask whether the model’s distribution over plural classes resembles human response distributions, and whether model uncertainty shifts under different conditions (e.g., from real nouns to nonsense nouns).

We study this question using a German Wug test dataset with human plural judgments on real, alternation, and nonce items from Molloy (2018). We compare three computational baselines: (i) a trigram model over local orthographic cues, (ii) gBERT, and (iii) mBERT, both probed in a zero-shot masked language modelling setup. We evaluate their predictions against empirical human response distributions using Jensen–Shannon distance (JSD), and we analyse uncertainty using entropy, varentropy, and an entropy-derived confidence measure.

Our research questions are:

1. How similar are model plural choice distributions to human response distributions across real, alternation, and nonce items?
2. Do models adapt their uncertainty across lexi-

cal conditions in a human-like way?

3. Are systematic model deviations consistent with local/subword heuristics rather than broader cue integration?

This paper makes four contributions: First, we provide a direct distributional comparison of human and model plural predictions for German nouns across lexical conditions. Second, we show that uncertainty behaviour (not only accuracy) distinguishes human and model processing profiles. Third, we demonstrate that a simple trigram baseline remains a strong comparator for German plural prediction. Fourth, we provide diagnostic evidence that many gBERT predictions are high-confidence and consistent with subword/lexical attraction effects.

2 Background and Related Work

German plural morphology has long served as a testing ground for theories of morphological productivity, including through Wug test paradigms (Berko, 1958) – where participants are tasked with pluralising a set of singular nonce nouns (i.e., novel, zero-frequency items) presented in sentence frames – to probe generalisation beyond memorised forms (e.g., Molloy (2018); Köpcke (1998)). A central finding in this literature is that plural selection is not arbitrary – it depends on a complex integration of phonological cues (e.g., rhyme and prosody) and grammatical cues, most notably noun gender. Furthermore, human responses often reflect graded preferences rather than categorical, deterministic choices.

In NLP, morphological processing has evolved from rule-based approaches and n-gram models to subword-based neural architectures (Baxi and Bhatt, 2024). Transformer language models with subword tokenisation improve lexical coverage by eliminating the “unknown” (<UNK>) token and have been widely used for probing grammatical knowledge (Graichen et al., 2026; López-Otal et al., 2025), including through the Wug paradigm (Anh et al., 2024; Samir and Silfverberg, 2022). While Transformer-based models achieve SOTA performance on high-resource data, they often struggle to generalise morphosyntactic rules to unseen stems, seemingly due to copy bias or a language’s morphological complexity itself (Liu and Hulden, 2021; Anh et al., 2024). However, the extent to which Transformer models exhibit human-like morphological generalisation remains unclear.

Most standard evaluations of the prediction of language phenomena rely on top-1 accuracy or related categorical measures. However, in ambiguous morphological systems, such metrics can hide important differences between human and model behaviour, particularly regarding how uncertainty is calibrated. We therefore treat plural prediction as a distributional task, utilising Jensen-Shannon Distance and entropy-based metrics to explicitly compare the prediction distributions and uncertainty profiles of BERT models against human empirical data.

3 Data and Task Definition

3.1 Task formulation

We formulate German plural prediction as a distribution prediction task. For each stimulus noun, the target is the empirical human response distribution over plural classes. Responses are mapped to five suffix classes:

$$\{-n, -e, -s, -r, -\emptyset\}.$$

Umlaut is not modelled as a separate class because BERT’s subword tokenisation strategies frequently struggle with this type of non-concatenative morphology (Wardini, 2022), complicating the isolation of umlaut as an independent feature.

This formulation captures graded human preferences and allows comparison of uncertainty across conditions.

3.2 Human plural judgment data

We use a German plural judgment dataset ($n = 9,198$) based on a Wug test design with human responses to real, alternation, and nonce nouns from Molloy (2018). The dataset contains responses from $n = 219$ speakers, across 42 target nouns, with items balanced across lexical conditions and grammatical gender. The three conditions are:

- **Real:** existing German nouns with one codified plural form;
- **Alternation:** existing German nouns found with both a zero and -s plural;
- **Nonce:** novel nouns requiring productive generalisation manipulated across phonological typicality.

Human responses are normalised and mapped to the five suffix classes.

3.3 Corpus data for trigram baseline

To construct a local-form baseline, we extracted noun plural patterns from the Leipzig German Web Corpus (2021 release, 1M documents).¹ After cleaning and normalisation, we isolated sentences containing plural nouns via the spacy POS tagger and a number of heuristic filters (e.g., German language, total length, proportion of digits), yielding 617,308 valid utterances. We extracted plural instances and mapped them to the same suffix-class inventory as the human data. The resulting dataset contained 1,165,162 plural instances across 175,587 unique lemmas.

Plural class distribution across the corpus was as expected, with the *-n* class dominating (656,734 instances) and the remaining instances distributed across the other classes.²

The trigram baseline uses the final trigram (orthographic ending) of each stimulus noun as a cue and estimates a plural-class distribution from the corpus counts. This provides a simple but linguistically relevant comparator as German plural choice is related to lexical typicality (Wunderlich, 1999), a feature which has previously been successfully modelled via final trigrams (Divjak et al., 2021).

4 Models and Experimental Setup

4.1 Trigram baseline

We estimate a maximum-likelihood trigram model over word-final trigrams and plural classes using Laplace smoothing to handle unseen sequences. For each stimulus noun, the model returns a probability distribution over plural classes conditioned on the noun-final trigram. This baseline tests the limits of orthographic information without contextualised representations.

4.2 Transformer models: gBERT and mBERT

We compare two masked language models: monolingual **German BERT** (gBERT, deepset/gbert-base) and **multilingual BERT** (mBERT, bert-base-multilingual-cased). Both models are used in a zero-shot probing setup

using the sentence frames previously presented to humans as prompts.

For each stimulus, we construct a masked prompt and extract plural-class evidence from model logits. The mask token is appended directly to the lemma without a space in-between (e.g., *Tafel*[MASK]) to probe suffix completion. To address tokenisation alignment issues where the mask might be merged, we use a fallback extraction rule reading the logit vector at the position immediately preceding {[SEP]} when the tokeniser output does not align with the intended mask position. We then map model outputs to the same plural-class inventory used for the human and trigram analyses.

4.3 Diagnostic analyses

We analyse disagreements between models and inspect high-confidence model predictions on nonce items. This tests whether BERT’s decision-making reflects random noise or systematic biases such as local subword/lexical attraction.

5 Evaluation Metrics and Statistical Analysis

5.1 Distributional similarity

We compare model distributions to human distributions using **Jensen–Shannon distance** (JSD), which takes the weighted average of two Kullback–Leibler divergences (KL). Let P denote the human distribution and Q the model distribution over the plural classes. JSD is defined as:

$$\text{JSD}(P, Q) = \frac{1}{2}\text{KL}(P|M) + \frac{1}{2}\text{KL}(Q|M),$$

$$M = \frac{1}{2}(P + Q),$$

where lower values indicate greater similarity to human plural-choice distributions.

5.2 Uncertainty metrics

To quantify uncertainty, we compute Shannon **entropy**:

$$H(P) = - \sum_{i=1}^K p_i \log_2 p_i,$$

where $K = 5$ is the number of plural classes. Higher entropy indicates a flatter, more uncertain distribution.

We also compute **varentropy** (variance of surprisal) as the probability-weighted variance of information content across plural classes: first computing surprisal as $-\log_2(p_i)$, then entropy as the

¹<https://wortschatz-leipzig.de/de/download/deu>

²Class distribution: *-n* (56%), *-ø* (19%), *-e* (18%), *-s* (5%), and *-r* (1%). Versus Penke and Krause (2002): most frequent *n > e > ø > r > s* least frequent. Since *-s* is replacing *-ø* (Salmons, 2018), its frequency is expected to have increased since 2002. *-r* is no longer productive, so a low frequency is expected. Under Wiese’s framework of prosodic morphology (Wiese, 1996), *-e* and *-ø* can be considered the same morpheme (allomorphs).

expected surprisal, and finally varentropy as the expected squared deviation of surprisal from entropy:

$$vH(P) = \sum_{i=1}^K p_i (-\log_2 p_i - H(P))^2.$$

Finally, we compute an entropy-derived **confidence** score:

$$C(P) = 1 - \frac{H(P)}{\log_2 K},$$

where confidence approaches 0 for a near-uniform distribution.

5.3 Statistical analysis

We analyse item-level metrics (JSD, entropy, confidence, and varentropy) across models and lexical conditions. For group comparisons, we use ANOVA and post-hoc Tukey tests, along with pairwise tests for targeted contrasts (e.g., human real vs. nonce uncertainty). Unless otherwise stated, significance was assessed at $\alpha = 0.05$.

6 Experiments

We run four experiments. All code can be found in the GitHub repository.³

E1: Distributional similarity to humans (RQ1)

We compare trigram, gBERT, and mBERT using JSD to human response distributions for each item, analysed separately for real, alternation, and nonce conditions.

E2: Uncertainty calibration across conditions (RQ2)

We compare entropy, varentropy, and confidence for humans and models across the three lexical conditions. The key question is whether model uncertainty increases from real to nonce items in a human-like way.

E3: Diagnostic disagreement analysis (RQ3)

We compare gBERT and trigram predictions to identify high-confidence disagreements, especially on nonce items, and inspect representative cases to test for local subword/lexical attraction effects.

E4: Feature dependence / variance decomposition (RQ3)

We analyse how variation in model and human behaviour is associated with lexical condition, grammatical gender, and trigram cue information.

7 Results & Discussion

Humans adapt uncertainty strongly across lexical conditions

Human plural-choice distributions show a clear uncertainty gradient across conditions (Figure 1). Mean entropy is lowest for real nouns ($H = 0.039$), increases for alternation items ($H = 0.799$), and is highest for nonce nouns ($H = 1.295$). The corresponding entropy-derived confidence decreases from real (0.983) to alternation (0.656) to nonce (0.442). This contrast is significant for all metrics across all conditions (Kruskal–Wallis H-test, $p < 0.005$). This indicates that human participants strongly calibrate uncertainty to lexical novelty and morphological ambiguity.

BERT shows weaker condition sensitivity than humans

In contrast to humans, both Transformer models show weaker adaptation of uncertainty across lexical conditions. Neither gBERT nor mBERT exhibits a comparably strong and statistically robust uncertainty shift on the main entropy/confidence contrasts. This suggests that the models do not track lexical novelty in the same way as human participants; they exhibit a different processing profile. This supports the idea that human plural judgments reflect metacognitive adaptation to lexical support, while the models rely on more stable internal heuristics across conditions.

The two BERT variants also differ from each other. gBERT tends to produce sharper, more peaked distributions (lower entropy / higher confidence), while mBERT exhibits a higher entropy floor and flatter distributions overall. This yields different failure modes: gBERT often produces sharper predictions and high-confidence overrides, which is problematic when the model is confidently wrong. By contrast, mBERT appears noisier and tends to distribute probability mass more diffusely, which may reflect weaker language-specific specialisation in a multilingual parameter space. mBERT consistently exhibits the highest divergence from humans across all three conditions.

A simple trigram baseline is a strong comparator

Mean JSD from human plural-choice distributions is shown in Table 1. The trigram baseline is unexpectedly competitive, outperforming gBERT in the real and nonce conditions, while gBERT performs more similarly to humans in the

³https://github.com/emmamolloy/BERT_morphology_generalisation

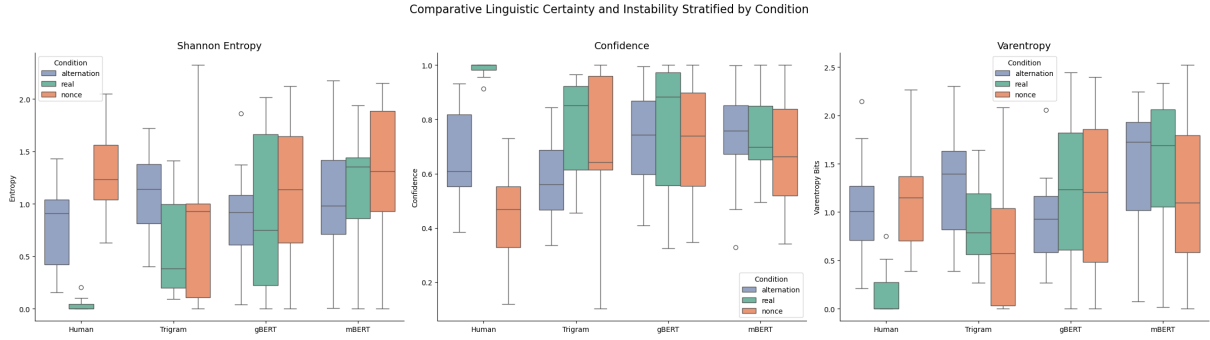


Figure 1: Uncertainty by lexical condition for humans, trigram, gBERT, and mBERT. Humans show a strong condition-sensitive shift (real → nonce), while BERT models show weaker condition sensitivity.

Condition	gBERT	mBERT	Trigram
Real	0.5937	0.8028	0.2961
Alternation	0.4496	0.5803	0.5622
Nonce	0.5963	0.6071	0.4845

Table 1: Mean JSD across conditions. Lower values indicate greater similarity to human distributions.

alternation condition. The trigram model’s failure in the alternations is due to an overapplication of the global maximum *-n*. This suggests that local orthographic cues capture a substantial portion of plural-choice behaviour in German. This could be due to orthography inadvertently also capturing prosodic shape, which is known to modulate German plural forms (Wiese, 1996). This means that improvements over *n*-gram baselines should not be assumed, and that any claim of human-like morphological generalisation should be evaluated against both strong local baselines and human uncertainty profiles.

BERT makes limited use of syntactic cues

We find limited evidence of functional gender sensitivity in gBERT under this setup. In a probing task where the syntactic gender of the determiner was switched, gBERT’s suffix switch rate was 0%. (Template: {gender} {lemma}, mehrere {lemma}[MASK].) Although attention maps indicate that the model attends to the determiner token, this syntactic information never overrides the local morphological cues derived from the stem. While mBERT’s uncertainty stays flat regardless of attention paid to the determiner containing gender information, gBERT becomes more confident as more attention weight is assigned to the determiner (Figure 3), suggesting that it does account for syntactic information to some extent. However, switching gender did not result in switching of plural class,

suggesting that local cues carry far more weight. This is supported by attention-map visualisations, where most attention was concentrated at the end of each sentence; after this, the most attention was generally assigned to the token immediately before the mask. Our experiment is consistent with a tendency to prioritise terminal local subword anchors over syntactic cues under this probing setup.

gBERT errors are systematic and often high-confidence gBERT’s deviations from the trigram baseline are not random. In our diagnostic comparison, gBERT and trigram disagree on 102/126 item-condition predictions (81%), and many disagreements occur at high confidence. This indicates that gBERT is not merely uncertain when it departs from local cues; rather, it often overrides them with a strong alternative preference.

Qualitative inspection of nonce cases suggests that these overrides are frequently consistent with local subword or lexical-attraction effects. For example, the nonce item *Kniempe* receives a high-confidence zero-plural prediction from gBERT, which appears consistent with attraction to known lexical/subword neighbours (e.g., *Knie*, zero plural) than with the trigram-based plural pattern. We also observe that many high-confidence overrides involve tokenisation fragmentation. Together, these patterns support the view that gBERT relies heavily on local subword anchors and lexical similarity when explicit morphological evidence is weak.

This points towards a plausible mechanism for gBERT’s deviations: local subword/lexical attraction. When a stimulus lacks strong lexical support from the training data, the model may fall back on subword fragments or nearest lexical analogies rather than integrating broader grammatical cues. This interpretation is consistent with the high-confidence disagreement cases and the observed

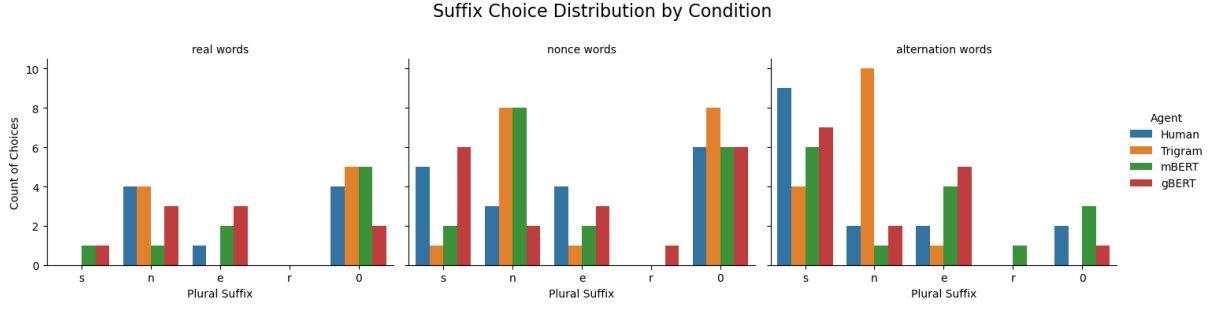


Figure 2: Both gBERT and mBERT select the non-productive *-r*. Trigram overapplies *-n* in the *s*-zero alternations. Under Wiese’s prosodic morphology framework, *-e* also counts as a zero ending.

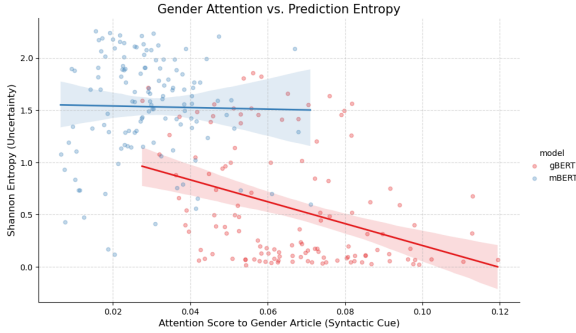


Figure 3: mBERT shows no change in entropy when paying attention to the article, while more attention reduces the entropy for gBERT.

association with tokenisation fragmentation. At the same time, these diagnostics are suggestive rather than conclusive: attention patterns and tokenisation behaviour do not by themselves establish causal mechanisms.

8 Limitations

This study has several limitations. First, the stimulus set is relatively small, which limits statistical power and the granularity of subgroup analyses. Second, our label mapping collapses plural behaviour into a five-class suffix inventory, which simplifies phenomena such as umlaut and may obscure finer distinctions in human and model preferences. Third, the BERT probing setup depends on a specific prompt and mask-position extraction strategy; tokenisation and fallback handling likely influence the resulting distributions. Fourth, our analyses are specific to German and to masked language model probing, so the findings should not be generalised automatically to other languages or model classes. These limitations do not undermine the main comparison, but they motivate more targeted follow-up experiments with larger datasets, alternative probing methods, and explicit cue-ablation

designs.

9 Conclusion

We compared human plural-choice distributions for German real, alternation, and nonce nouns with predictions from a trigram baseline, gBERT, and mBERT. Using JSD and uncertainty metrics, we found that humans strongly calibrate uncertainty to lexical novelty, while BERT models show weaker condition sensitivity. We also found that a simple trigram baseline is highly competitive, and that many gBERT deviations are high-confidence and consistent with local subword/lexical attraction rather than human-like cue integration. These results highlight the value of distributional and uncertainty-based evaluation for morphological modelling and suggest that future work should test whether character-aware or morphology-aware models can better match human uncertainty calibration. Overall, our findings argue for evaluating morphological competence in language models using distributional and uncertainty-based measures, not only top-1 accuracy. A model may appear competent under categorical scoring while still diverging substantially from human cue integration and uncertainty calibration.

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A Group Contributions

This was an individual project. The author was responsible for data preprocessing, model setup and probing, statistical analysis, visualisations, and writing.

B AI Statement

ChatGPT 5.2 was used for generating draft phrasing of selected sections, proofreading, debugging, and creating plots. It was not used to generate or run the experiments, and it did not produce the underlying data, code, or statistical outputs. All analyses, figures, and reported results were executed and verified by the author.

C Data & Software Metadata

Table 2: Corpus data metadata (Leipzig corpus).

D1: Data licence	
Corpus data:	Leipzig Corpora Collection (German Web Corpus, 2021 release), CC BY
D2: Dataset name and properties	
Name:	Leipzig Corpora Collection (German Web Corpus)
Creator:	University of Leipzig (Wortschatz Leipzig project)
Format:	UTF-8 encoded, tab-separated text files (multiple files; e.g., sentences.txt, words.txt, optional POS-tagged sentence/word lists, source lists, metadata)
Fields:	Varies by file type; examples include: Sentence_ID, Sentence, Word_ID, Word, Frequency
Temporal coverage:	2021 release
Data quality:	Large-scale web crawl corpus; project-provided standardised format and encoding. In this study, additional cleaning/filtering and normalisation were applied before extraction of plural instances (see notebook).
D3: Additional dataset notes	
Representativeness:	Broad web-derived German text, suitable for estimating high-level distributional tendencies at scale; includes heterogeneous sources rather than a single genre.
Limitations:	Web text is not balanced for genre/register/dialect and may contain noise, duplicates, OCR-like artefacts, tagging errors, and non-standard spellings. POS tagging and heuristic filtering can introduce additional errors/bias. Corpus frequency patterns may differ from psycholinguistic elicitation data and from lexical resources (e.g., CELEX).

Table 3: Human judgment data metadata.

D1: Data licence	
Human judgment data:	Research dataset, not publicly licensed (author-created)
D2: Dataset name and properties	
Name:	German plural judgment dataset
Creator:	Emma Molloy, collected for a BA thesis at the University of Cambridge. In the original thesis, the data were reported descriptively (percentage proportions) and were not analysed using statistical or geospatial methods.
Format:	CSV, 219 rows $\times$ 49 columns (7 respondent-information columns, 42 item-response columns)
Fields:	speaker_id, demographics (e.g., age, gender), dialect affiliation(s), 33 test-item responses, 9 filler responses
Temporal coverage:	Spring 2018
Data quality:	Responses manually normalised; < 3% missingness; additional response mapping applied for plural-class analysis in the present study.
D3: Additional dataset notes	
Representativeness:	Demographics: Majority monolingual female respondents aged 19–30; most reported spending < 1 year abroad. Skewed toward Austrian and Bavarian dialect speakers.
Limitations:	No postcode/GPS precision; subjective self-report of dialect affiliation; elicited judgments may differ from naturalistic production; plural responses are collapsed into a reduced suffix-class inventory for modelling comparisons.

Table 4: Software metadata.

S1: Current software version	Python 3.10.18, JupyterLab 4.0
S2: GitHub repo	https://github.com/emmamolloy/BERT_morphology_generalisation
S3: Legal software licence	None
S4: Computing platform / OS	macOS 15.4.1
S5: Installation requirements / dependencies	All dependencies listed in environment.txt in repo
S6: Special software	None
S7: Additional information	None